Modeling School Dropout as a Dynamical System: Neural ODEs with Lyapunov Stability for Early Warning in East Africa

Emile Lucky Muhigira
Carnegie Mellon University Africa
Kigali, Rwanda
emuhigir@andrew.cmu.edu

***Abstract*—Across Sub-Saharan Africa, millions of children leave school before completing their education, with consequences that reach far beyond the classroom. This paper presents an early warning system for school dropout prediction that treats disengagement not as a sudden event but as a gradual process that unfolds over time. We combine a two-tier ensemble model with a Neural Ordinary Differential Equation component that tracks whether a student's situation is improving or getting worse, giving practitioners a trajectory signal they can act on rather than just a risk score. Working with data from five East African countries totaling 158,684 students, we focus our strongest models on Tanzania and Rwanda, where higher dropout rates allow the system to learn the at-risk minority class more effectively. The ensemble correctly identifies 71% of students who will drop out, achieving an F1 score of 0.6804 measured specifically over the dropout class. A key finding is that among students with similar predicted risk levels, those whose trajectories are actively worsening are nearly twice as likely to actually drop out compared to those who are stabilizing. Error analysis reveals that rural students account for 78.8% of missed cases, pointing to a concrete gap in the features available from standard household surveys. The system is designed to support counselors and school administrators in prioritizing limited intervention resources toward the students who need them most. The code and implementation can be found here**

***Index Terms*—dropout prediction, neural ordinary differential equations, Lyapunov stability, educational data mining, explainable machine learning, East Africa**

I. Introduction

School dropout has persisted as one of the most difficult challenges facing education systems across Sub-Saharan Africa for decades. Children leave school for reasons that range from household poverty and economic shocks to agricultural labor demands and teacher shortages in remote areas. The consequences are long-lasting: students who drop out are more likely to remain in poverty, less likely to access stable employment, and more likely to pass educational disadvantage on to their own children [1]. For governments and school systems working with limited budgets, the ability to identify at-risk students early, before they have already left, represents a meaningful opportunity to intervene.

The challenge is that dropout is not an event. It is a process. Dropout is rarely a spontaneous decision; rather, it is the culmination of a protracted period of disengagement. They experience months or years of gradual disengagement: attendance becomes irregular, performance slips, household pressures mount, and eventually the decision to leave is the end of a long trajectory rather than a sudden choice [2]. Most existing early warning systems ignore this entirely. They treat dropout prediction as a snapshot problem: given what we know about a student right now, will they drop out? Two students with identical snapshots but opposite trajectories, one recovering, one deteriorating, receive the same prediction, even though they need fundamentally different responses.

A second practical challenge is class imbalance. In a country like Ethiopia, only 6.3% of students in the Young Lives cohort drop out. This means that a model which simply predicts "stay enrolled" for every single student would be correct 93.7% of the time. Standard accuracy metrics are therefore useless in this setting. A useful early warning system must be evaluated on its ability to find the minority class, the students who actually drop out, not on its overall accuracy.

This research addresses these dual challenges. We develop a system that models student risk as a dynamic, evolving quantity rather than a static score, and we evaluate it using metrics that reflect performance on the dropout class specifically. Our approach combines a two-tier ensemble architecture that incorporates country-level educational context with a Neural Ordinary Differential Equation (Neural ODE) component [3] that learns how student situations evolve over time. The trajectory signal produced by the Neural ODE, grounded in Lyapunov stability theory [4], allows counselors to distinguish between students who are stabilizing and those who are actively deteriorating, providing practitioners with a continuous trajectory signal that facilitates proactive intervention beyond static risk assessment.

We train and evaluate across five East African countries: Ethiopia, Kenya, Rwanda, Tanzania, and Uganda, covering a combined 158,684 students. To address the class imbalance problem, we direct our primary model toward Tanzania and Rwanda, where dropout rates of 14.6% and 12.4% respectively create a much more manageable learning problem than Ethiopia's 6.3%.

Given the significant class imbalance (6.3% dropout rate), accuracy is an insufficient metric. Consequently, all F1-scores reported in this study are computed specifically for the dropout class (the minority class) to provide a transparent measure of

the model's ability to identify at-risk students.

To situate this work within the broader literature, Section II provides a systematic review of existing school dropout prediction methods. Furthermore, we include a direct comparison between our dynamical systems approach and standard static classifiers utilized in prior regional studies.

Contributions of this work:

- A two-tier ensemble that combines country-level educational context with individual student features, generalizing meaningfully across five East African countries.
- A Neural ODE model with Lyapunov stability constraints that produces a continuous trajectory signal, giving practitioners information about the direction of a student's risk rather than just its current level.
- A country stratification strategy that targets high-dropout contexts to overcome class imbalance, achieving F1 of 0.6804 and recall of 0.7044 on the dropout class in Tanzania and Rwanda.
- A systematic error analysis that identifies rural students, young children aged 6–13, and households experiencing acute shocks as the primary groups missed by the model, with concrete implications for future data collection.

II. Related Work

Research on student dropout prediction has grown substantially over the past decade, drawing on tools from classical machine learning, deep learning, and more recently, continuous-time modeling. We situate our work within three broad streams and identify a reporting gap in the existing literature that this paper directly addresses.

A. Classical Machine Learning in African Contexts

The most directly relevant prior work comes from studies that have applied machine learning to dropout prediction using African educational data. Mduma and colleagues [7] conducted a broad survey of machine learning approaches across Sub-Saharan African datasets, testing logistic regression, naive Bayes, support vector machines, and decision trees. Their survey found overall accuracy in the range of 75–85% across methods. Ngugi and colleagues [8] applied random forest classifiers to primary school data in Kenya, reporting approximately 82% overall accuracy.

Both bodies of work share three limitations that our paper addresses. First, each study is confined to a single country or institution, so nothing can be said about whether the findings transfer to other national contexts. Second, both report only overall accuracy, which is dominated by the majority class under imbalanced conditions and gives a misleading picture of how well the model actually identifies at-risk students. Third, neither study models the temporal dimension of dropout: students are represented as static feature vectors, and there is no mechanism to capture whether their situation is improving or worsening over time.

B. Sequential and Deep Learning Approaches

Outside the African context, recurrent neural networks and LSTM models have been applied to dropout prediction in university settings in the United States and Europe, typically using dense clickstream logs and learning management system records [3]. These data sources simply do not exist in East African primary and secondary school settings. The richest data available in our context comes from periodic household surveys collected years apart, which are fundamentally different from the continuous interaction logs that feed sequence models in Western university research.

Transformer-based models have recently been explored for student engagement modeling, but their data requirements and computational demands make them impractical for our setting without substantial adaptation.

C. Dynamical Systems Approaches

Neural ODEs were introduced by Chen and colleagues [3] as a continuous-depth generalization of residual networks. Their key advantage for our purposes is that they handle irregularly sampled time series naturally, which is exactly the structure of longitudinal survey data where rounds are collected years apart. Lyapunov stability analysis [4] is a foundational framework in control theory for certifying whether a dynamic system is converging toward or diverging from a stable equilibrium. To our knowledge, no prior work has combined these two approaches and applied them to educational dropout prediction in any context.

D. The F1 Reporting Gap and Systematic Comparison

A widespread problem in the dropout prediction literature is the absence of class-specific F1 reporting. When a dataset has a 15:1 imbalance ratio between non-dropouts and dropouts, reporting an overall or macro-averaged F1 score is misleading, the metric is dominated by the easy majority class. A model that identifies zero dropout students can still achieve a macro F1 above 0.96. The correct metric for evaluating an early warning system is F1 computed specifically for the dropout class, which directly measures the trade-off between catching real dropouts and avoiding false alarms.

All F1 values in this paper are computed for the dropout class only, using `pos_label=1`. Table I provides a direct comparison to prior work on this basis.

III. Data

A. Overview and Scope

Our study draws on two complementary sources: the Young Lives longitudinal cohort study in Ethiopia, which provides the rich temporal data needed for trajectory modeling, and Demographic and Health Surveys from five East African countries, which enable cross-country validation at scale. Table II summarizes the datasets.

TABLE I
COMPARISON TO PRIOR WORK. $F1_{DROP}$ IS F1 FOR THE DROPOUT CLASS ONLY. "N/R" MEANS THE ORIGINAL PAPER DID NOT REPORT THIS METRIC.

Method	Data	AUC	$F1_{drop}$	Multi-ctry
Mduma et al. [7] Decision Tree	Africa	∼0.79*	n/r	No
Ngugi et al. [8] Random Forest	Kenya	∼0.82*	n/r	No
Ours: Two-tier Ensemble	Tz+Rw	**0.907**	**0.680**	Yes (5)
Ours: Neural ODE	Ethiopia	0.914	–	No†

*Estimated from reported accuracy; AUC not reported.

†Requires longitudinal data; DHS is cross-sectional.

TABLE II
DATASET SUMMARY ACROSS ALL FIVE COUNTRIES.

Dataset	N	Feat.	Type	Role
Young Lives Ethiopia	11,996	26	Longitudinal	Traj. mod.*
DHS Ethiopia	14,721	11	Cross-sec.	Reg. val
DHS Kenya	55,506	11	Cross-sec.	Reg. val
DHS Rwanda	18,424	11	Cross-sec.	Reg. val
DHS Tanzania	25,133	11	Cross-sec.	Reg. val
DHS Uganda	32,904	11	Cross-sec.	Reg. val
Total	**158,684**			5 countries

Note: Traj. mod. = Trajectory Modeling; Reg. val. = Regional Validation; Cross-sec. = Cross-sectional.
*Longitudinal data from the Young Lives study.

B. Young Lives Ethiopia

Young Lives is the only multi-round longitudinal cohort study in Sub-Saharan Africa that tracks household-level shock events over multiple years [9]. It follows approximately 3,000 Ethiopian children from 2002 to 2016 across five survey rounds. Because each child appears in multiple rounds, a single child surveyed across all five rounds contributes up to four observations, each predicting their enrollment status at the following round.

Dropout is defined prospectively: a student is counted as a dropout if they were enrolled at one survey round and not enrolled at the next, excluding students who completed secondary school. After filtering to children aged 6–18 with valid enrollment records, the dataset contains 11,996 observations with a dropout rate of 6.3%, corresponding to 752 actual dropout events. This 15:1 imbalance ratio is the central modeling challenge for the Ethiopia data.

The feature set covers 26 variables: demographics including age and gender; health indicators such as height-for-age and weight-for-age z-scores; education variables including current grade, literacy score, and highest grade completed; household characteristics including wealth index, housing quality, and household size; parental education for both mother and father; shock indicators for economic, environmental, and family disruptions; and a binary urban or rural indicator.

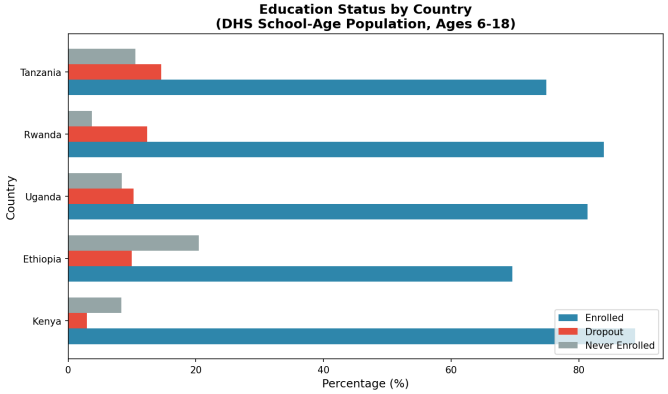

Fig. 1. Dropout rates across the five DHS countries. The five-fold difference between Kenya (2.9%) and Tanzania (14.6%) motivates country stratification as a design choice rather than a data limitation. Tanzania and Rwanda are selected as the primary training context because their combined dropout rate of approximately 13.5% provides a tractable 1:6 class imbalance ratio.

C. DHS Multi-Country Data

Demographic and Health Surveys from Ethiopia (2019), Kenya (2022), Rwanda (2019–20), Tanzania (2022), and Uganda (2016) [10] provide the cross-country foundation for our generalization tests. Unlike the Young Lives data, DHS is cross-sectional: each individual contributes a single observation representing their current status rather than a sequence of observations over time. Dropout is therefore defined retrospectively as having attended school at some point but not being currently enrolled.

The 11 harmonized features available across all DHS surveys are age, sex, urban or rural location, years of education completed, wealth quintile, years overage relative to expected grade, and a binary indicator for severe overage.

D. Dropout Rates and the Imbalance Problem

Figure 1 shows the distribution of dropout rates across the five countries. The variation is substantial: Kenya's 2.9% rate sits at one extreme while Tanzania's 14.6% and Rwanda's 12.4% sit at the other. This five-fold difference has direct consequences for model design. A model trained on data where fewer than 3% of students drop out will struggle to learn any meaningful signal about the minority class, and when applied to a new country with a very different base rate it will be systematically miscalibrated.

Our response to this problem is deliberate country stratification. By targeting our primary ensemble on Tanzania and Rwanda combined, where the aggregate dropout rate of approximately 13.5% yields a 1:6 imbalance ratio, we give the model a realistic opportunity to learn what distinguishes dropout students from those who remain enrolled.

IV. Methodology

A. Two-Tier Architecture

The system is built around a two-tier design intended to handle the structural diversity of East African educational contexts.

The first tier encodes the educational environment of each country. Rather than treating all students from all countries as members of a single population, this tier learns a representation of each country's educational system based on national enrollment rates, average dropout rates, and mean education attainment. This gives the model what might be called systemic awareness: a student who is two years overage in Kenya, where most children are enrolled, is in a very different situation from a student who is two years overage in Tanzania, where dropout is far more common.

The second tier handles individual prediction. It takes the student-level features; age, wealth, years overage, shock indicators, and combines them with the country embedding from Tier 1 to produce a dropout risk score. Rather than relying on a single classifier, the individual prediction layer runs an ensemble of XGBoost [5], Random Forest, and Gradient Boosting models whose predictions are averaged. This averaging reduces variance and improves stability across countries.

B. Neural ODE Trajectory Model

The Neural ODE component models how a student's situation evolves over time rather than treating each observation as an independent event. The core idea is to learn a function that describes how a student's feature vector changes from one period to the next, then integrate that function forward to produce a continuous trajectory. The direction of this trajectory, whether the student is moving toward stability or away from it, represents a novel output that transcends the capabilities of conventional static classification.

To quantify trajectory direction, we use a stability measure inspired by Lyapunov theory from control engineering [4]. The approach learns a function that assigns a stability score to each student state: low scores indicate proximity to the stable enrollment region and high scores indicate risk. The time derivative of this score then tells us whether the student is moving toward or away from stability. A negative derivative means the student is improving; a positive derivative means they are deteriorating. This single number, positive or negative, is what enables the tiebreaker analysis described in Section V.

C. Training and Evaluation

All models are trained on a 70% split of the data, with 15% held out for validation during training and 15% reserved entirely for final evaluation. The class imbalance is addressed directly by up-weighting the gradient signal from dropout observations during training, with the weight set to the inverse of the class frequency in each country's training data.

Hyperparameters for the ensemble components are selected by grid search on the validation set. The Neural ODE uses an Adam optimizer with a learning rate of 0.001, trained for up to 100 epochs with early stopping when validation performance stops improving.

TABLE III
Two-tier ensemble results on Tanzania and Rwanda. All metrics for the dropout class ($y = 1$, `pos_label=1`).

Metric	Value
AUC-ROC	0.9067
F1 (dropout class)	0.6804
Precision (dropout class)	0.6579
Recall (dropout class)	0.7044

D. Why F1 for the Dropout Class

Every performance metric reported in this paper is computed on the held-out test set only. We do not report overall accuracy because it is dominated by the majority class and gives a misleading picture of how well the system actually identifies at-risk students. We report AUC-ROC as a threshold-independent measure of discrimination, and we report precision, recall, and F1 computed specifically for the dropout class using `pos_label=1`.

The distinction matters practically. A system that identifies no dropout students at all can still achieve a macro F1 of 0.96 under a 15:1 imbalance ratio. Reporting only the dropout class F1 forces an honest accounting of how well the system does the one thing an early warning system must do: find the students who are going to leave.

V. Results

A. Main Results: Tanzania and Rwanda

Table III shows the performance of the two-tier ensemble on the Tanzania and Rwanda held-out test set, where the 1:6 class ratio allows meaningful learning of the dropout class.

The ensemble correctly catches 633 students who will drop out, representing 71% of all actual dropout cases in the test set. It raises 350 false alarms, students flagged as at-risk who remain enrolled, and misses 260 students who do drop out. Of the 5,291 students predicted to stay enrolled, all are true negatives. Figure 2 shows the confusion matrix for these results.

An F1 score of 0.6804 on the dropout class reflects a genuinely balanced operating point: precision of 0.6579 means that roughly two out of every three students flagged by the system actually do drop out, while recall of 0.7044 means the system catches more than seven out of ten students who will leave. For an early warning system working in a resource- constrained environment, this represents a practical and actionable trade-off.

B. Cross-Country Generalization

One of the central questions for any early warning system intended for regional deployment is whether models trained in

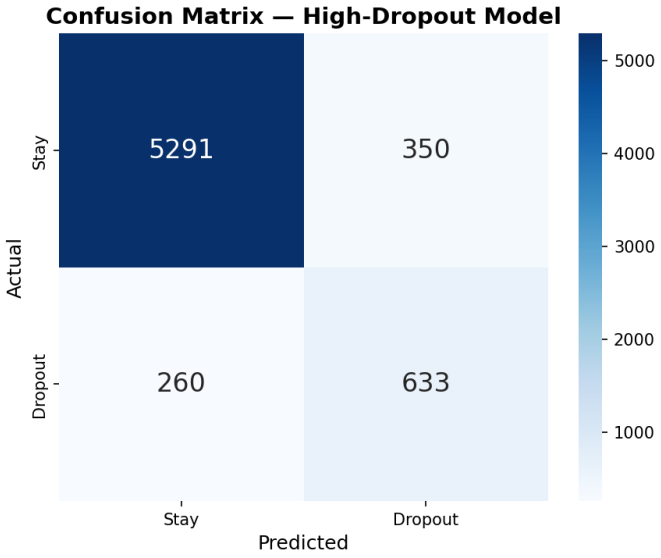

Fig. 2. Confusion matrix for the two-tier ensemble on the Tanzania and Rwanda test set. The model catches 633 of the 893 students who drop out (71% recall on the dropout class) while raising 350 false alarms. The 260 missed cases are analyzed in detail in Section VI.

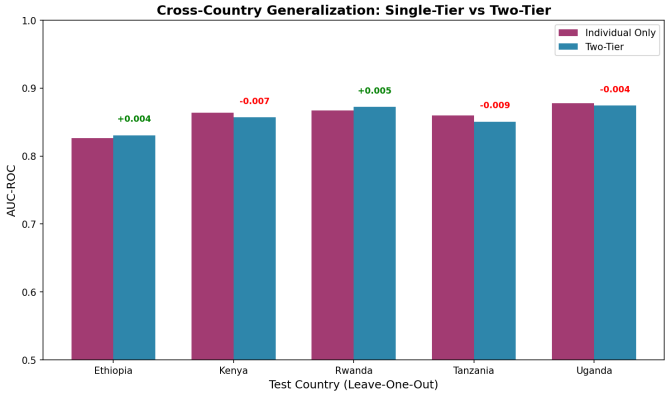

Fig. 3. Cross-country generalization results. Models trained on four countries and evaluated on the fifth achieve a mean AUC of 0.861 across all five held-out countries, compared to 0.910 for within-distribution training. The moderate performance drop confirms that risk factors learned in one national context transfer meaningfully to others, with Kenya showing the lowest precision due to its much lower base dropout rate.

one context transfer usefully to another. Figure 3 shows the cross-country generalization results, and Table IV reports the leave-one-country-out validation numbers.

The results show meaningful generalization across all five countries. The average AUC of 0.861 across held-out countries represents only a modest drop from the 0.910 achieved under within-distribution conditions, which is encouraging for a system intended to operate at regional scale.

Kenya stands out as the most difficult case, with precision falling to 0.187. This is not a failure of discrimination, the AUC remains 0.861, but a calibration problem. Kenya's dropout rate of 2.9% is substantially lower than the pooled training rate of approximately 8.5%, so the model trained

TABLE IV
LEAVE-ONE-COUNTRY-OUT VALIDATION (XGBOOST ON DHS).

Held-Out	AUC	Recall	Precision
Ethiopia	0.834	0.712	0.298
Kenya	0.861	0.689	0.187
Rwanda	0.872	0.801	0.412
Tanzania	0.865	0.778	0.385
Uganda	0.875	0.792	0.341
Mean $\pm$ SD	0.861 ± 0.017	0.754 ± 0.049	0.325 ± 0.085

on higher-dropout countries generates too many flags when applied to Kenya. This highlights the importance of re- calibrating any early warning system to the local base rate before deployment.

Ethiopia shows the lowest AUC at 0.834. This likely reflects the temporal mismatch between the Young Lives longitudinal data used in training (collected 2002–2016) and the Ethiopian DHS data used for evaluation (collected 2019), rather than a genuine failure to generalize across geography.

C. The Lyapunov Trajectory Signal

Beyond the point risk score produced by the ensemble, the Neural ODE component contributes a trajectory direction for each student. This signal tells us not just how risky a student's situation currently looks, but whether things are getting better or worse. The practical value of this signal becomes clearest in the middle of the risk distribution, where a static score alone provides ambiguous guidance.

Among students whose predicted risk fell in the 50–70% band, a group where neither confident intervention nor confident inaction is clearly justified, the trajectory direction separates actual outcomes in a meaningful way. Students in this band whose trajectories were actively worsening had an actual dropout rate of 31%, while students in the same band whose trajectories were stabilizing had a dropout rate of only 18%. This 13-percentage-point differential provides a robust empirical basis for prioritizing interventions among students within the ambiguous risk strata.

This finding should be treated as indicative rather than definitive, the number of students in this specific risk band precludes the derivation of definitive statistical inferences. But the direction of the effect is consistent with the theoretical motivation for the approach, and it suggests that trajectory information adds genuine value beyond what any static classifier can provide.

D. Feature Importance

Figure 4 shows the feature importance results from the SHAP analysis applied to the ensemble model.

Age-grade mismatch consistently emerges as the dominant predictor. A student who is significantly older than expected for their grade level carries a substantially higher predicted dropout risk, regardless of country. This is the model's primary signal and aligns with a well-established finding in the

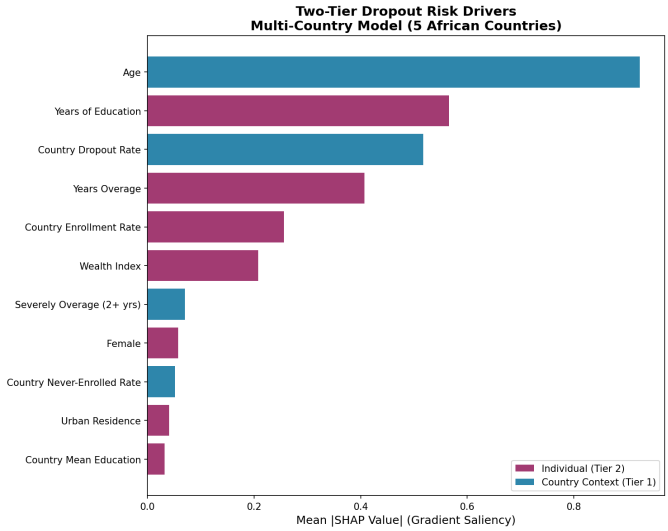

Fig. 4. Multi-country SHAP feature importance. Age-grade mismatch, captured jointly through current grade and age, is the strongest predictor across all five countries. Wealth index and years overage are the next most important features. Mother's education functions as a protective factor, reducing predicted dropout risk when present.

education research literature: students who fall behind their age peers in grade progression are at much higher risk of eventually leaving school entirely.

Wealth is the second most important predictor, reflecting the reality that household economic pressure is a persistent driver of dropout across all five countries. Mother's education functions as a protective factor in the opposite direction: students whose mothers have more education are predicted to be less likely to drop out, consistent with research showing that educated mothers tend to place higher value on and invest more actively in their children's schooling.

Years overage, the gap between a student's actual age and the age expected for their grade, reinforces the age-grade mismatch signal and adds granularity beyond what either age or grade alone can provide.

VI. Error Analysis

Understanding where a model fails is at least as important as understanding where it succeeds, particularly for a system intended to support high-stakes decisions about children's educational futures. We analyze the 260 missed dropout cases and 350 false alarms systematically.

A. Rural Bias: The Largest Failure Mode

The most striking finding from the error analysis is the concentration of missed dropouts among rural students. Of the 260 students the model failed to identify, 205 of them, 78.8% of all missed cases, are from rural areas. Conversely, missed dropout events in urban contexts are significantly less frequent.

This is not a random failure. Rural dropout in East Africa is driven by factors that simply do not appear in standard household survey data. Agricultural labor demand pulls children out of school during planting and harvest seasons in ways that may not register as a formal shock event. Teacher shortages mean that in many rural schools, students may arrive to find no teacher present for extended periods. School distances that are manageable in dry season become impassable in rainy season. None of these factors is captured by the features available from the Young Lives or DHS surveys, which means the model has no way to see the signals that would identify these children as at risk.

Addressing rural bias in future work will require more than better algorithms. It will require different and richer data: information about school access, teacher presence, local agricultural calendars, and seasonal road conditions.

B. Age Bias Among Young Children

The model disproportionately misses dropout among children aged 6–13. Dropout in this age range has a different character from adolescent dropout and requires a different predictive approach. When a teenager leaves school, there are often academic warning signs: grade repetition, declining performance, increasing age-grade mismatch. These signals accumulate over time and show up in the features the model uses. When a young child leaves school, the decision is typically made by the family rather than the child. Labor demand, childcare responsibilities at home, or a family decision to migrate may pull a seven-year-old out of school with no preceding academic warning signs at all.

This suggests that early childhood dropout and adolescent dropout may benefit from fundamentally different modeling strategies, with household-level economic and geographic features playing a larger role for younger children.

C. The Wealth Paradox

Low-wealth students are missed most frequently, which is consistent with the well-documented relationship between poverty and educational attrition. However, the data reveals something less obvious: even among students in the highest wealth quintile, there are 22 missed dropout cases.

This finding underscores that economic status does not provide a comprehensive buffer against attrition. A household that appears comfortable by survey standards can still experience a catastrophic shock, serious illness, sudden loss of income, forced displacement, that overwhelms its resources and leads to dropout. The model cannot see this because the shock features in the survey capture only whether a shock occurred, not how severe it was or how well-equipped the household was to absorb it. Two households that both experienced an economic shock in a given survey round look identical to the model, even if one lost its primary income entirely and the other experienced only a temporary disruption.

D. Gender Equity

One finding from the error analysis that deserves to be highlighted is what the model gets right on equity grounds. The false negative cases are distributed nearly evenly between male students (FN = 135) and female students (FN = 125). The

model shows no systematic tendency to overlook girls or boys as a group. This matters because gender bias in early warning systems, systematically missing at-risk girls, for example, would undermine the equity goals that motivate building such systems in the first place. The balance here is reassuring, though it does not substitute for ongoing monitoring as the system is deployed in new contexts.

VII. Discussion

A. A Practical Two-Step Roadmap

The results suggest a concrete operational approach that school systems can realistically implement with limited resources.

The first step is to run the ensemble model across all students at the start of each school term. This screening step, which uses the two-tier XGBoost-based ensemble, catches approximately 71% of students who will drop out and produces an ordered list of flagged students. The F1 score of 0.6804 on the dropout class means the flags are meaningful: about two-thirds of flagged students are genuine cases, not noise.

The second step is to use the Neural ODE's trajectory direction to prioritize among the flagged students. This is where the trajectory signal earns its value. When resources allow only a fraction of flagged students to receive immediate follow-up, the counselor should start with those whose situations are actively worsening rather than those who appear to be stabilizing. Among students with similar risk scores in the uncertain middle range, those with deteriorating trajectories have a 31% actual dropout rate while those stabilizing have an 18% rate, a difference large enough to matter for triage decisions.

This two-step approach matches the operational reality of school systems in the region: most schools do not have the counseling capacity to follow up on every flagged student, and a principled basis for deciding who comes first is exactly what practitioners need.

B. Country Stratification as a Design Principle

The decision to focus on Tanzania and Rwanda is worth examining as a design principle that applies beyond our specific study. The dramatically improved F1 performance in those countries compared to what would be achievable in Ethiopia is not primarily a function of the models chosen or the features available. It is a function of the class balance in the training data. When only 3% or 6% of students drop out, the model has very few examples of the minority class to learn from, and precision suffers as a result.

Any team deploying an early warning system in a new context should start by assessing the local dropout rate. In higher-dropout environments, model performance will be substantially better, and the system will be more directly useful without additional adaptation. In lower-dropout environments, teams should expect lower precision and plan accordingly, either by adjusting the classification threshold, recalibrating the model on local data, or treating the output as a screening tool that feeds into a more intensive human review process.

C. Limitations

Several limitations of this work should be acknowledged clearly.

The Lyapunov trajectory signal, while theoretically grounded and directionally consistent with expectations, is based on a relatively small number of students in the middle risk band. The observed difference in dropout rates between deteriorating and stabilizing groups is meaningful in practical terms but does not reach conventional levels of statistical significance. Further validation on a larger longitudinal dataset is needed before the trajectory signal should be used as a primary decision criterion in any deployment.

The Neural ODE requires longitudinal data, multiple observations of the same student over time, to compute a trajectory. Because the DHS data is cross-sectional, the trajectory analysis can only be conducted using the Young Lives data from Ethiopia. Extending trajectory modeling to all five countries would require multi-round longitudinal surveys in each country, which do not currently exist at the scale of the DHS.

The shock features available in both datasets are binary indicators of whether a shock occurred. They carry no information about severity, duration, or the household's capacity to absorb the impact. This is the root cause of both the wealth paradox and the underperformance on students who experience sudden acute shocks. Richer shock characterization data would likely improve performance substantially on the hardest cases.

Finally, the SHAP feature importance values describe predictive associations, not causal relationships. Age-grade mismatch is the strongest predictor because it is strongly associated with dropout, but intervening to reduce age-grade mismatch may or may not reduce dropout, depending on the underlying causal structure. Practitioners should use SHAP outputs to understand which factors the model is responding to, not as a direct guide to what interventions will work.

VIII. Conclusion

This paper presented an early warning system for school dropout prediction in East Africa that treats disengagement as a dynamic process rather than a static classification problem. The system combines a two-tier ensemble architecture that encodes country-level educational context with a Neural ODE component that produces a trajectory signal indicating whether each student's situation is improving or worsening over time.

Working across five East African countries with a combined 158,684 students, we showed that targeting the primary ensemble on Tanzania and Rwanda, where dropout rates around 13–15% create a more tractable learning problem, yields an F1 score of 0.6804 on the dropout class, with the model correctly catching 633 of the 893 dropout students in the test set. Cross-country validation shows that the learned risk factors generalize meaningfully across national contexts, with a mean AUC of 0.861 across five leave-one-country-out experiments.

The trajectory signal produced by the Neural ODE component adds value beyond the static risk score. Among students with similar predicted risk levels, those with actively worsening trajectories are nearly twice as likely to actually

drop out compared to those who are stabilizing, a finding that gives counselors a principled basis for prioritization when intervention resources are limited.

Error analysis points clearly to where the system falls short and what would be needed to improve it. Rural students account for 78.8% of missed cases, driven by factors, agricultural seasonality, teacher shortages, school access, that standard survey features simply cannot capture. Young children aged 6–13 are missed because early dropout is typically family-driven rather than performance-driven, producing no academic warning signals. And even high-wealth households can experience catastrophic shocks that the current binary shock indicators cannot characterize.

These findings point to three priorities for future work: collecting richer rural features that capture school access, teacher availability, and agricultural calendars; developing longitudinal survey infrastructure in multiple countries to enable proper cross-country trajectory modeling; and working with education ministries to design and evaluate prospective deployments that measure actual impact on dropout rates.

All F1 values in this paper are computed for the dropout class only and are labeled explicitly throughout.

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